MATIAS AHRENSDORF

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Data and software engineer who takes public data end to end, sourcing it (FOIL, IPUMS, Census, NIBRS, HUD), modeling it (Python, fixed-effects panel regression, PCA, ML imputation), and shipping it (Snowflake/Tableau pipelines and published writing). Research support for nationally recognized scholars; 11 bylines in the *Wall Street Journal*, *City Journal*, and *RealClear*.

EXPERIENCE

Manhattan Institute — *Data & Software Engineer* New York, NY · Apr 2024 – Present

- Build end-to-end research pipelines for scholars and fellows, sourcing and modeling federal and municipal data (IPUMS, Census/NHGIS, NIBRS, HUD, CDC, SEER, GSS) into analysis-ready panels; work has supported a TED talk, a published book, and multiple institute reports.
- Ran fixed-effects panel regressions on a **74,000-row county-by-year panel** for Roland Fryer's TED talk on hip-hop's cultural influence, with clustered standard errors and survey weighting; **re-specified the exposure variable to be standardized within each subset**, making effect sizes comparable across race and age cohorts, and rebuilt the regression code into a single parameterized function with built-in robustness checks.
- Consolidated **3.4M 311 service requests across 15 of the 25 largest U.S. cities** — each with an incompatible schema — plus NIBRS petty-crime and HUD homelessness counts, into a unified public-disorder dataset for a senior fellow.
- Modeled judicial leniency across **454,000 NYC arraignment records**, using PCA and residualized regression to separate judge-specific behavior from case mix.
- Built **LLM integrations via MCP servers connecting Snowflake and Airtable**, plus reverse-ETL jobs feeding modeled data into custom CRM fields, replacing recurring manual pulls for the development and marketing teams.
- File FOIL requests on behalf of scholars pursuing government data, including a **two-year action against the MTA** that escalated to outside counsel and ended in full production of the requested dataset.
- Publish analysis and data visualizations in the *Wall Street Journal*, *City Journal*, *RealClear*, and *Compact*; manage content and releases across three web properties (manhattan.institute, City Journal, Adam Smith Society).

Compact Magazine — *Web Developer (Contract)* Remote · Nov 2025 – Present

- Maintain and update the publication's website on an ongoing retainer, implementing content and functionality changes for the editorial team and serving as the outside technical contact for the site.

LS Direct Marketing — *Financial Analyst* Suffern, NY · Jan 2023 – Apr 2024

- Built and automated the CFO's core reporting stack — sales tax, deferred revenue, COGS, commissions, and budget-vs-actuals — replacing manual Excel workflows with Python/Pandas scripts.
- Managed source databases and built validation checks reconciling reporting across departments, catching discrepancies before they reached leadership.
- Produced Tableau dashboards presented at board meetings, tracking company and client performance trends.

Compass Group — *Sr. Payroll Analyst* Charlotte, NC · Sep 2021 – Jan 2023

- Automated manual payroll processes and led the transition to paperless operations, eliminating recurring processing hours and reducing labor-cost leakage.
- Built KPI and trend reporting, and delivered training webinars translating payroll data issues into decisions for operations management.

SELECTED PUBLICATIONS

Wall Street Journal (opinion) · *City Journal* (4) · *RealClearMarkets / Defense / Books* (3) · *Compact* · *Washington Examiner* · Manhattan Institute report on indexing New York State taxes to inflation. Beats: Latin America policy and financial risk.

TECHNICAL SKILLS

Languages & Analysis: Python (pandas, NumPy, scikit-learn, statsmodels, linearmodels, pyfixest, geopandas), SQL, R, JavaScript
Methods: Fixed-effects panel regression, clustered standard errors, PCA, random forests, KNN imputation, geospatial analysis, survey microdata (IPUMS, GSS), FOIA/FOIL research
Data & Platform: Snowflake, Tableau, reverse ETL, MCP servers / LLM integration, Airtable, Zapier, AWS, MongoDB, React/Node.js, WordPress
Business: Financial reporting automation, forecasting, data validation, board reporting
Other: Spanish (bilingual) · Microsoft Excel (MOS Certified)

EDUCATION

Elon University — B.S. Business Administration, Finance; Minor, Spanish Language & Literature 2020
Presidential Scholar · National Hispanic Honor Society · Study abroad: Barcelona, Spain; Japan & South Korea

University of North Carolina at Charlotte — Full-Stack Web Development Bootcamp 2022